Question 1:

Solve the equation $x^4 - 3x^2 + 2 = 0$.

a) Let $u = x^2 \quad \Rightarrow \quad (u)^2 = (x^2)^2 \quad \Rightarrow \quad u^2 =$ ___?___.

b) Write the equation $x^4 - 3x^2 + 2 = 0$ in terms of u : ___?___.

c) The solutions that satisfy the equation $x^4 - 3x^2 + 2 = 0$ are ___?___.

Question 2:

Solve the equation $x^4 - 6x^2 - 7 = 0$

a) Let $u = x^2 \quad \Rightarrow \quad (u)^2 = (x^2)^2 \quad \Rightarrow \quad u^2 =$ ___?___.

b) Write the equation $x^4 - 6x^2 - 7 = 0$ in terms of u: ___?___.

c) The solutions that satisfy the equation $x^4 - 6x^2 - 7 = 0$ are ___?___.

Question 3:

Solve the equation $x - 4\sqrt{x} + 3 = 0$

a) Let $u = \sqrt{x} \quad \Rightarrow \quad (u)^2 = \left(\sqrt{x}\right)^2 \quad \Rightarrow \quad u^2 =$ ___?___.

b) Write the equation $x - 4\sqrt{x} + 3 = 0$ in terms of u: ___?___.

c) The solutions that satisfy the equation $x - 4\sqrt{x} + 3 = 0$ are ___?___.

Question 4:

Solve the equation $x^6 + 7x^3 - 8 = 0$

a) Let $u = x^3 \quad \Rightarrow \quad (u)^2 = (x^3)^2 \quad \Rightarrow \quad u^2 =$ ___?___

b) Write the equation $x^6 + 7x^3 - 8 = 0$ in terms of u: ___?___.

c) The solutions that satisfy the equation $x^6 + 7x^3 - 8 = 0$ are ___?___.